

Maternal Health Risk Stratification

Short Analysis Summary

.Atefe .Asadi | StudyBuild | September 2026

Objective

This summary compares the three maternal risk groups and evaluates a simple multinomial Logistic Regression model. The analysis focuses on class specific performance, especially errors involving High Risk observations, rather than judging the model by accuracy alone.

Data review

- The project file contains 1,014 observations, seven variables, and no missing values.
- The initial review found 562 exact repeated rows.
- Two observations with HeartRate equal to 7 bpm were excluded because the values are physiologically implausible for this dataset.
- Exact repeated rows were removed for the primary analysis, leaving 451 observations. Because patient identifiers are unavailable, this is reported as a methodological assumption.
- Thirty five identical predictor profiles are linked to more than one RiskLevel label. These observations were kept because the disagreement may reflect missing clinical information or uncertainty in the original labels.

Main findings

Finding	Result
Largest High Risk standardized difference	Blood Sugar, absolute SMD = 1.61
Next largest standardized differences	SystolicBP = 0.70; DiastolicBP = 0.61
Primary model	Multinomial Logistic Regression with StandardScaler
Primary split	80/20 stratified split; random state = 42
Primary accuracy	0.670
Primary High Risk recall	0.565, 13 of 23 identified
Primary Mid Risk recall	0.048, 1 of 21 identified
Most common error	Mid Risk predicted as Low Risk, 17 cases
Missed High Risk error	High Risk predicted as Low Risk, 10 cases
Largest High Risk coefficient	BS = +1.176
Grouped five fold accuracy	0.607 ± 0.029
Grouped five fold Macro F1	0.593 ± 0.032
Grouped five fold High Risk recall	0.695 ± 0.057
Grouped five fold Mid Risk recall	0.330 ± 0.139

Interpretation

Blood Sugar has the largest standardized difference for High Risk and the largest positive coefficient in the fitted High Risk model. The primary Logistic Regression model has 67.0% accuracy, but it identifies only 56.5% of High Risk observations and 4.8% of Mid Risk observations. All ten missed High Risk observations were assigned to Low

Risk. Their mean BS was 7.35 mmol/L, compared with 12.81 mmol/L among High Risk observations that were identified correctly.

Sensitivity analysis

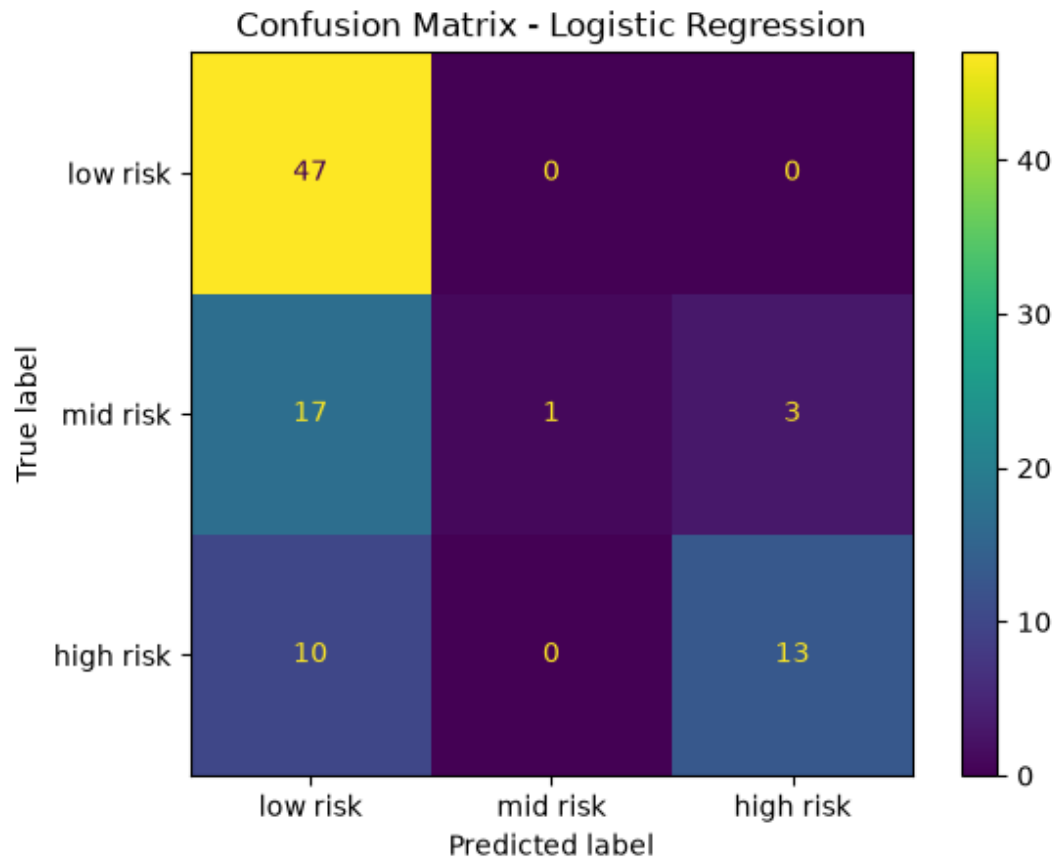
A five fold grouped analysis retained the repeated observations while keeping identical predictor profiles within one fold. Mean accuracy was 0.607, Macro F1 was 0.593, High Risk recall was 0.695, and Mid Risk recall was 0.330. No identical predictor profile appeared in both training and test data in any fold.

Limitations

- The primary modeling sample is small and contains only six predictors.
- Patient identifiers are unavailable, so repeated rows cannot be verified at the person level.
- Some identical predictor profiles have different RiskLevel labels.
- The primary result uses one held out test set. The grouped analysis is a sensitivity check, not external validation.
- The reported associations and coefficients should not be interpreted as causal clinical effects.

Conclusion

This project provides a transparent statistical baseline for maternal health risk classification. Its main value lies in the class specific evaluation, inspection of missed High Risk observations, and direct examination of how repeated rows affect the performance estimates. The model is not intended for diagnosis or clinical decision making.

Key visual

Confusion matrix for the primary Logistic Regression model. The largest error counts are Mid Risk predicted as Low Risk and High Risk predicted as Low Risk.